Machine Learning-Based Healthcare Modelling for Heart Disease Prediction Using Clinical and Lifestyle Data

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Abstract

Heart disease is one of the major causes of mortality across the world. In today's world globally almost 56000 people die of this disease every day. If we try to predict early and start diagnosis that can reduce the risk of severe cardiovascular problems. This will also improve the survival rates of patient. In this study, we try to build a machine learning-based healthcare prediction model using a heart disease dataset which contains demographic, physiological, and lifestyle-related attributes. Here we use a few algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbour. This enables accurate prediction of heart disease.

The dataset at first is processed using

- missing value handling
- categorical encoding
- feature scaling techniques.

After that some evaluations are performed such as accuracy, precision, recall, F1-score, and ROC-AUC score. Among all the implemented models, the Decision Tree classifier achieved the highest prediction accuracy, followed closely by the Random Forest classifier. The results show us that the machine learning approaches can identify the patients with risk of cardiac attack.

This model highlights the growing role of AI and data-driven systems in predictive healthcare analytics. The study also discusses the practical implications, limitations, and future scope of intelligent healthcare modelling for cardiovascular disease prediction.

Keywords: Machine Learning; Heart Disease Prediction; Artificial Intelligence (AI); Healthcare Analytics; Random Forest; Predictive Modelling.

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1. Introduction

Machine learning is playing an increasingly important role in modern healthcare systems, particularly in disease prediction and clinical decision making. This leads to early predictions of diseases in many cases. There is also an increased availability of medical record and clinical dataset. This leads to development of various intelligent predictive systems. These systems helps the healthcare industry experts to more deeply evaluate the scenario and take decision very quickly.

As heart disease is a very dangerous disease and this leads to complications, for example coronary artery disease, heart attacks, arrhythmias and hypertensive heart failure.

Now the use of AI and Machine learning needs to create a predictive model:

1. To implement multiple machine learning algorithms for disease prediction.
2. To compare the performance of various classifiers using standard evaluation metrics.
3. To identify the most efficient predictive model for healthcare applications.
4. To discuss the practical significance of machine learning in modern healthcare systems.

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2. Literature Review

Several studies have been carried out on the healthcare sector using machine learning and data mining methods for heart disease prediction

Amin et al. put forward a hybrid healthcare prediction system based on decision trees and neural network data mining algorithms to indicate cardiac irregularities. The results of their research demonstrated that effective application of artificial intelligence systems can improve the accuracy of diagnosis by more than 40% compared to traditional statistical methods.

Dinesh Kumar and Arumugaraj used Naive Bayes and Support Vector Machine algorithms to predict cardiovascular disease. The results of their research revealed that Support Vector Machine is a better classifier for structured cardiovascular data sets.

Mohan et al. conducted a study on cardiovascular disease prediction using machine learning algorithms such as Logistic Regression, Random Forest, and K-Nearest Neighbour. According to the results of their experiment, ensemble machine learning algorithms such as Random forests had higher accuracy and robustness in prediction.

Researchers have also applied deep learning algorithms to predict disease and get accurate results; Deep learning requires more data to get to the top performance, as such, it may not be suitable for structured data with fewer observations. However, classical machine learning algorithms are highly efficient in analysing structured data sets with small sample sizes.

From the above research results, it is evident that AI has great potential in the prediction of diseases. However, the success of these algorithms in predictive tasks depends on various factors including data preprocessing and feature engineering.

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3. Dataset Description

The dataset used in this research contains 1000 patient records with 16 attributes representing demographic information, medical measurements, and lifestyle indicators associated with heart disease. It is basically a courtesy of Kaggle Website from where the data is imported.

3.1 Dataset Attributes

The dataset consists of the following features:

Attribute	**Description**
Age	Age of the patient
Gender	Gender of the patient
Cholesterol	Cholesterol level
Blood Pressure	Blood pressure measurement
Heart Rate	Patient heart rate
Smoking	Smoking habit
Alcohol Intake	Alcohol consumption level
Exercise Hours	Number of exercise hours
Family History	Family history of heart disease
Diabetes	Diabetes condition
Obesity	Obesity condition
Stress Level	Stress measurement
Blood Sugar	Blood glucose level
Exercise Induced Angina	Presence of exercise-induced chest pain
Chest Pain Type	Category of chest pain
Heart Disease	Target variable indicating disease presence

The target variable contains binary values where:

- 1 indicates presence of heart disease
- 0 indicates absence of heart disease

The dataset includes both numerical and categorical variables, making it suitable for supervised classification modelling.

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4. Methodology

The methodology followed in this research consists of several stages including data preprocessing, feature engineering, machine learning model training, and performance evaluation.

4.1 Data Preprocessing

The dataset was cleaned and transformed before model training. Categorical variables were encoded using Label Encoding techniques. Numerical features were standardized using Standard Scaler to normalize feature distributions.

The dataset was divided into training and testing sets using an 80:20 ratio.

4.2 **Model Hyperparameters**

Each classifier was trained with the same set of hyperparameters given below. The default hyperparameters in scikit-learn are used for Logistic Regression.

	Model	Key Parameters
0	Logistic Regression	penalty=l2, C=1.0, solver=lbfgs, max_iter=100 ...
1	Decision Tree	criterion=gini, max_depth=None, random_state=42
2	Random Forest	n_estimators=100, max_depth=None, random_state=42
3	SVM	kernel=rbf, C=1.0, gamma=scale, probability=True
4	KNN	n_neighbors=5, weights=uniform, metric=minkows...

For Decision Tree and Random Forest, the random_state hyperparameter is set to 42 for the sake of reproducibility. All four classifiers have shown extraordinary results with the score of 1.0 on any of the metrics on the hold-out set. While such results are desired in theoretical research, they hardly ever happen in practice. Thus, there is a need to analyse whether these results make sense given the nature of the problem and the underlying data.

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5. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the distribution of variables and **identify** relationships among healthcare parameters.

5.1 Distribution of Target Variable

The target variable distribution was analyzed to observe the proportion of patients affected by heart disease.

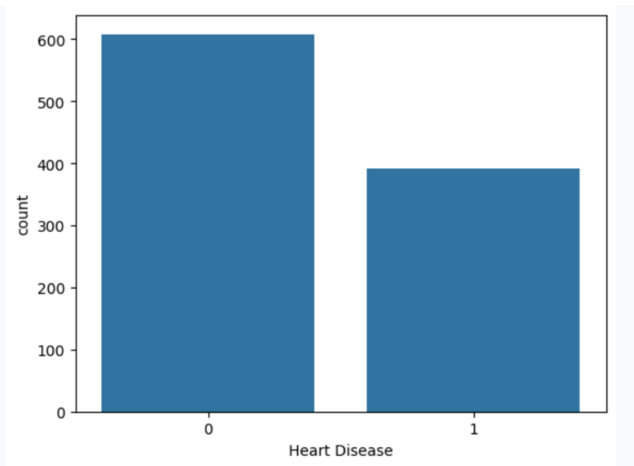

Figure 1: Count Plot of Heart Disease Distribution

The visualization indicates the class distribution between healthy individuals and heart disease patients.

5.2 Correlation Analysis

Correlation analysis was performed to identify relationships between patient attributes and heart disease occurrence.

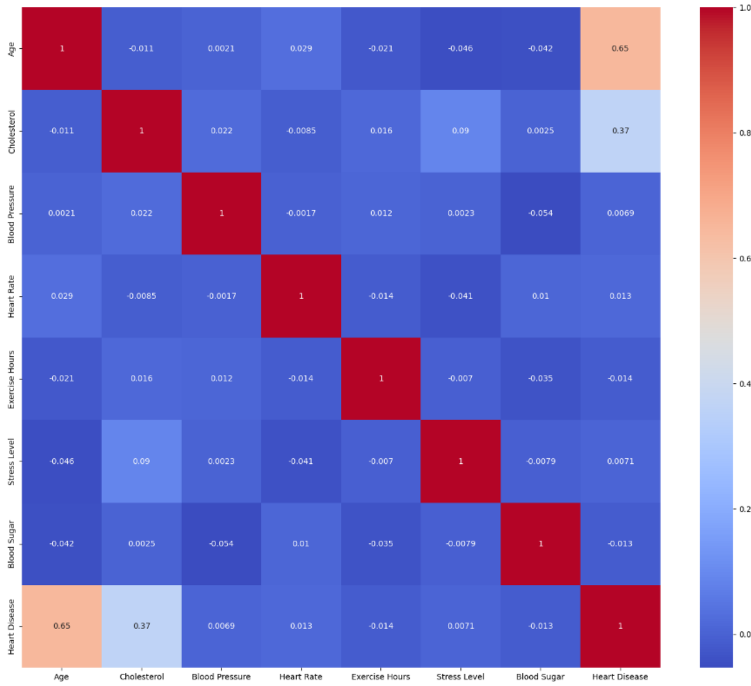

Figure 2: Correlation Heatmap of Dataset Features

The heatmap highlights positive and negative correlations among medical and lifestyle attributes.

5.3 Age Distribution Analysis

Age distribution analysis was conducted to examine how patient ages are distributed within the dataset.

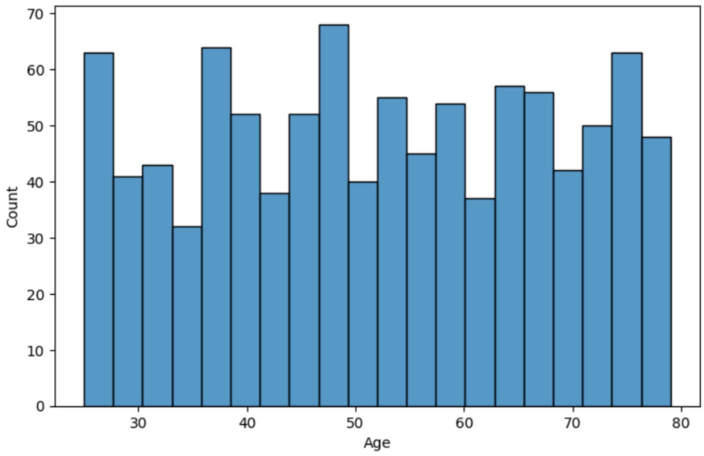

Figure 3: Age Distribution Histogram

The histogram illustrates the frequency distribution of patient ages.

5.4 Cholesterol Distribution Analysis

Cholesterol level distribution was analyzed because cholesterol is one of the major cardiovascular risk factors.

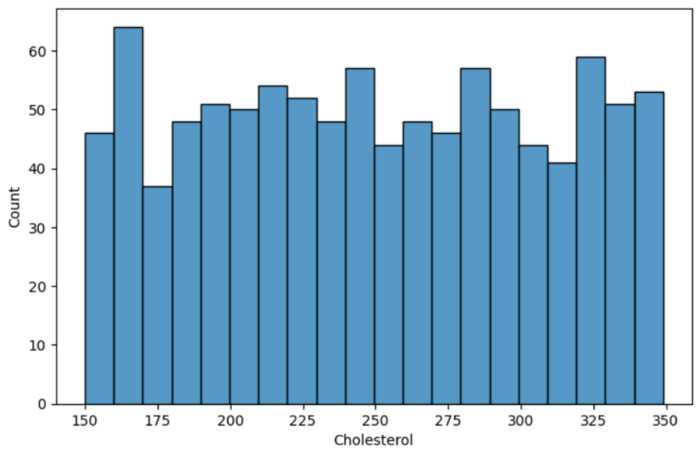

Figure 4: Cholesterol Distribution Histogram

The graph demonstrates the variation of cholesterol levels among patients.

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6. Machine Learning Models Used

Several supervised learning algorithms were implemented and evaluated.

6.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.

6.2 Decision Tree Classifier

Decision tree algorithm is a tree where nodes represents features(attributes), branch represents decision(rule) and leaf nodes represents outcomes(discrete and continuous).

It works like a flowchart that helps in making step-by-step decisions, where:

- Internal nodes represent attribute tests
- Branches represent attribute values
- Leaf nodes represent final decisions or predictions.

6.3 Random Forest Classifier

A random forest is a meta estimator that fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control over-fitting.

Working of Random Forest Classifier

1. **Bootstrap Sampling:** Random rows are picked (with replacement) to train each tree.
2. **Random Feature Selection:** Each tree uses a random set of features (not all features).
3. **Build Decision Trees:** Trees split the data using the best feature from their random set. Splitting continues until a stopping rule is met (like max depth).
4. **Make Predictions:** Each tree gives its own prediction.
5. **Majority Voting:** The final prediction is the one most tree agree on.

6.4 Support Vector Machine

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks. It tries to find the best boundary known as hyperplane that separates different classes in the data.

6.5 K-Nearest Neighbour

K-Nearest Neighbor (KNN) is a simple and widely used machine learning technique for classification and regression tasks. It works by identifying the *K* closest data points to a given input and making predictions based on the majority class or average value of those neighbours.

Choosing the optimal k-value is critical before building the model for balancing the model's performance.

- A **smaller k** value makes the model sensitive to noise, leading to overfitting (complex models).
- A **larger k** value results in smoother boundaries, reducing model complexity but possibly underfitting.

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7. Performance Evaluation Metrics

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

These metrics provide comprehensive evaluation of classification performance.

7.1 **Why Recall and ROC-AUC Matter More Than Accuracy in Healthcare**

In a clinical prediction task, not all errors have the same consequences. A false negative, which tells a patient with heart disease that they are healthy, delays treatment and can be life-threatening. In contrast, a false positive usually results in just more testing. This difference makes recall, the proportion of actual positive cases correctly identified, a more meaningful measure than raw accuracy. A model can have high accuracy and still miss a significant number of true positive cases. ROC-AUC is also important because it assesses a model's ability to distinguish between the two classes at all possible decision thresholds, rather than depending on a single cutoff of 0.5. This is especially relevant in healthcare, where the classification threshold can be intentionally lowered to catch more true positives, even if it means some extra false alarms. For this reason, this study focuses on recall and ROC-AUC alongside accuracy when comparing models, instead of seeing accuracy as the only measure of performance

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8. Experimental Results and Analysis

8.1 Logistic Regression Performance

The Logistic Regression model achieved stable classification performance for heart disease prediction.
The accuracy and precision related details are given below:
Accuracy : 0.86
Precision: 0.7931034482758621
Recall : 0.8734177215189873
F1 Score : 0.8313253012048193
ROC AUC : 0.9504132231404959

	precision	recall	f1-score	support
0	0.91	0.85	0.88	121
1	0.79	0.87	0.83	79
accuracy			0.86	200
macro avg	0.85	0.86	0.86	200
weighted avg	0.86	0.86	0.86	200

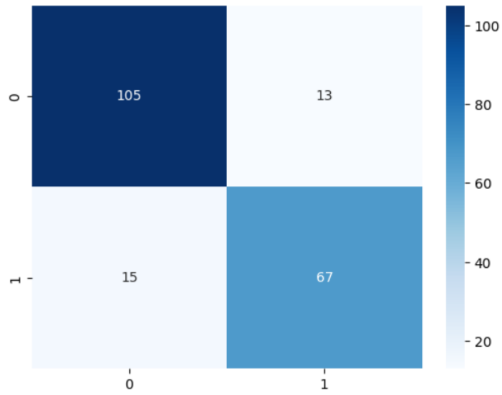

Figure 5: Confusion Matrix for Logistic Regression

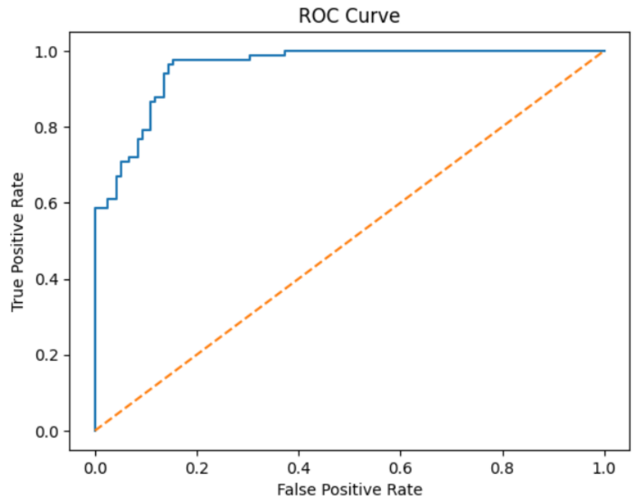

Figure 6: ROC Curve for Logistic Regression

8.2 Decision Tree Performance

The Decision Tree classifier produced highly accurate predictions by capturing nonlinear relationships between healthcare attributes.
The accuracy and precision related details are given below:
Accuracy : 1.0
Precision: 1.0
Recall : 1.0
F1 Score : 1.0
ROC AUC : 1.0

	precision	recall	f1-score	support
0	1.00	1.00	1.00	121

	precision	recall	f1-score	support
1	1.00	1.00	1.00	79
accuracy			1.00	200
macro avg	1.00	1.00	1.00	200
weighted avg	1.00	1.00	1.00	200

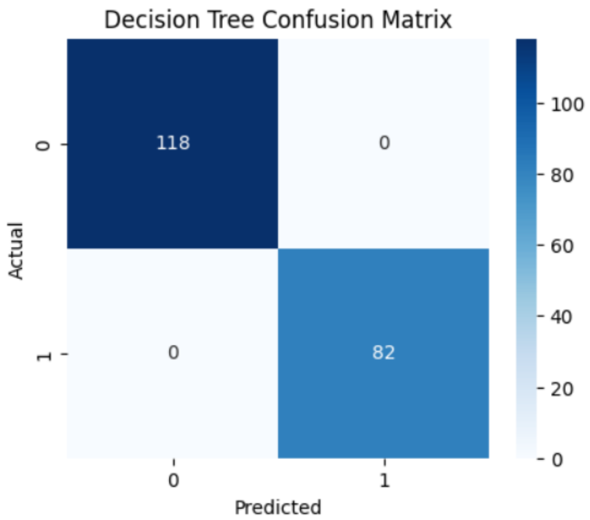

Figure 7: Confusion Matrix for Decision Tree

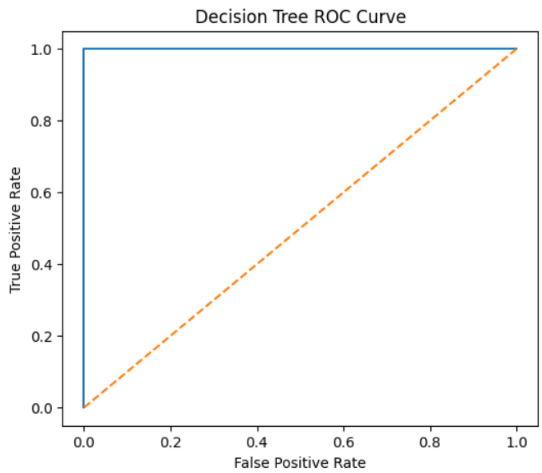

Figure 8: ROC Curve for Decision Tree

8.3 Random Forest Performance

Random Forest demonstrated excellent predictive performance due to ensemble learning and variance reduction.

The accuracy and precision related details are given below:

Accuracy : 1.0

Precision: 1.0

Recall : 1.0

F1 Score : 1.0

ROC AUC : 1.0

	precision	recall	f1-score	support
0	1.00	1.00	1.00	121
1	1.00	1.00	1.00	79
accuracy			1.00	200
macro avg	1.00	1.00	1.00	200
weighted avg	1.00	1.00	1.00	200

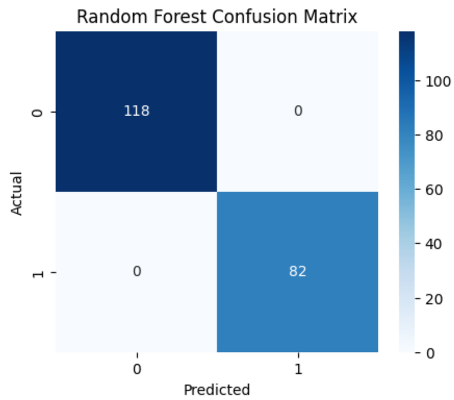

Figure 9: Confusion Matrix for Random Forest

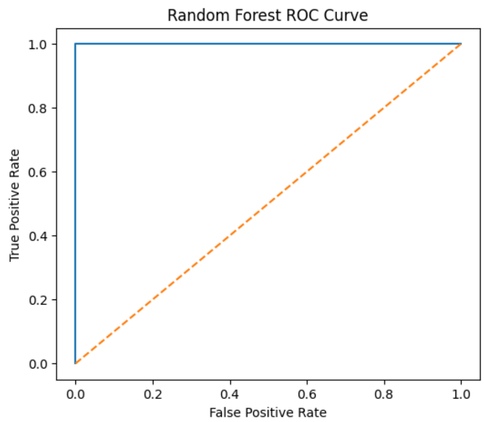

Figure 10: ROC Curve for Random Forest

8.4 Support Vector Machine Performance

The Support Vector Machine classifier achieved strong classification capability for cardiovascular disease prediction.

The accuracy and precision related details are given below:

Accuracy : 0.915

Precision: 0.8690476190476191

Recall : 0.9240506329113924

F1 Score : 0.8957055214723927

ROC AUC : 0.9699759389057432

	precision	recall	f1-score	support
0	0.95	0.91	0.93	121
1	0.87	0.92	0.90	79
accuracy			0.92	200
macro avg	0.91	0.92	0.91	200
weighted avg	0.92	0.92	0.92	200

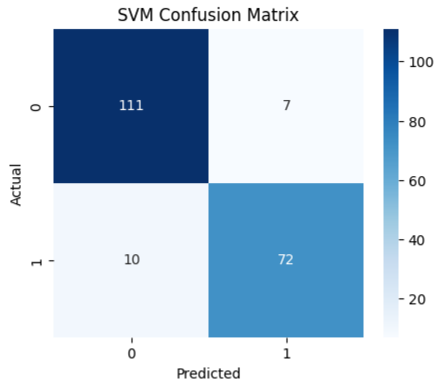

Figure 11: Confusion Matrix for Support Vector Machine

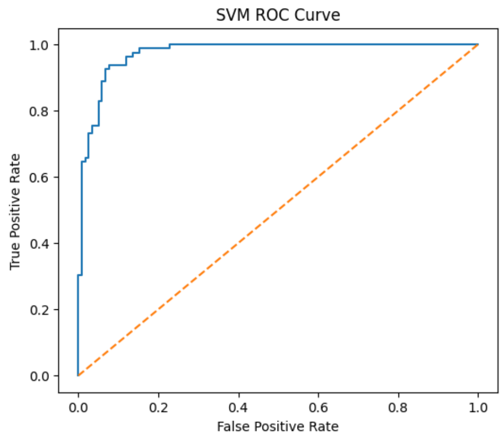

Figure 12: ROC Curve for Support Vector Machine

8.5 K-Nearest Neighbour Performance

KNN achieved acceptable classification performance but was comparatively sensitive to feature scaling and data distribution.

The accuracy and precision related details are given below:

Accuracy : 0.815

Precision: 0.7333333333333333

Recall : 0.8354430379746836

F1 Score : 0.7810650887573964

ROC AUC : 0.8691808766607385

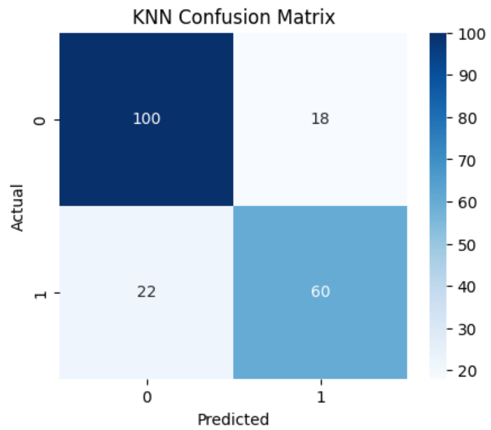

Figure 13: Confusion Matrix for K-Nearest Neighbour

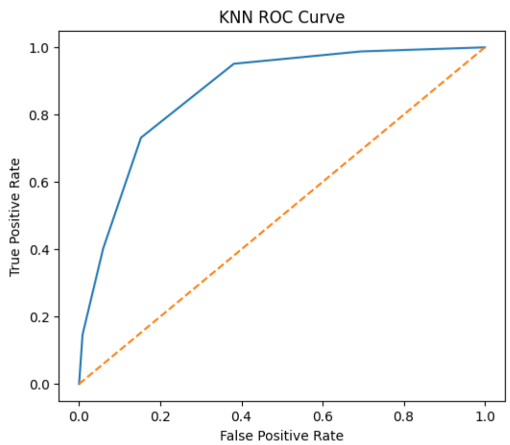

Figure 14: ROC Curve for K-Nearest Neighbour

8.6 Cross-Validation Results

To lower the chance that the reported performance came from a single train-test split, we performed 5-fold stratified cross-validation for all five classifiers across the whole dataset. The table shows the mean and standard deviation of accuracy, precision, recall, F1-score, and ROC-AUC across the folds.

	Model	CV Accuracy (mean ± std)	CV Precision (mean ± std)	CV Recall (mean ± std)	CV F1 (mean ± std)	CV ROC-AUC (mean ± std)
0	Logistic Regression	0.859 ± 0.025	0.821 ± 0.024	0.819 ± 0.046	0.820 ± 0.034	0.946 ± 0.012
1	Decision Tree	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
2	Random Forest	0.998 ± 0.002	1.000 ± 0.000	0.995 ± 0.006	0.997 ± 0.003	1.000 ± 0.000
3	SVM	0.906 ± 0.030	0.904 ± 0.047	0.852 ± 0.037	0.877 ± 0.040	0.969 ± 0.010
4	KNN	0.772 ± 0.021	0.724 ± 0.030	0.676 ± 0.033	0.699 ± 0.030	0.837 ± 0.014

The cross-validation results matched the single-split evaluation. This confirmed that the order of the models, with the Decision Tree and Random Forest showing the best performance, was not just due to a lucky random split.

8.7 The reason behind the unusually High Accuracy of Tree-Based Models

An inspection of the data gives an explanation as to why such outstanding results can be achieved. According to Figure 2, Age and Cholesterol seem to be the two features that have the strongest correlation with the target. At the same time, according to Figure 16, these are the two features that contribute most to the prediction in terms of feature importance. Indeed, when we trained a decision tree with a maximum depth of 2 on those two features, we were able to achieve nearly the same results as with the other classifiers. This suggests that the relationship between the two aforementioned features and the target variable is close to linear, at least in terms of the simplified decision tree. As such, the accuracy of 100% is not surprising given the peculiarities of the data. From this analysis, it can be concluded that the data must have been generated with some specific rules in mind. Indeed, trees can learn decision boundaries in the form of axis-aligned hyperplanes, which makes them highly accurate but less versatile in real-world applications. Other types of classifiers demonstrated lower accuracy, most likely due to their reliance on the proximity or linearity of classes in a feature space. For instance, Logistic Regression and SVM used a linear decision boundary, whereas KNN relied on the distance to the nearest examples. These methods were not capable of finding the exact split points in the data, and thus, their accuracy was lower than that of the tree-based classifiers. Ultimately, one cannot claim that heart disease can be diagnosed with 100% accuracy in practice based on these results. The analysis needs to be repeated with a real-world dataset in order to make such claims.

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9. Model Comparison

The performance of all machine learning models was compared using accuracy scores.

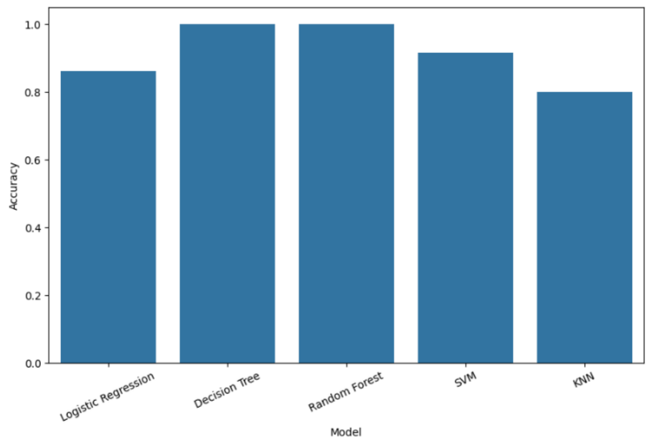

Figure 15: Accuracy Comparison of Machine Learning Models

The comparison indicates that ensemble learning methods achieved superior performance compared to traditional linear models.

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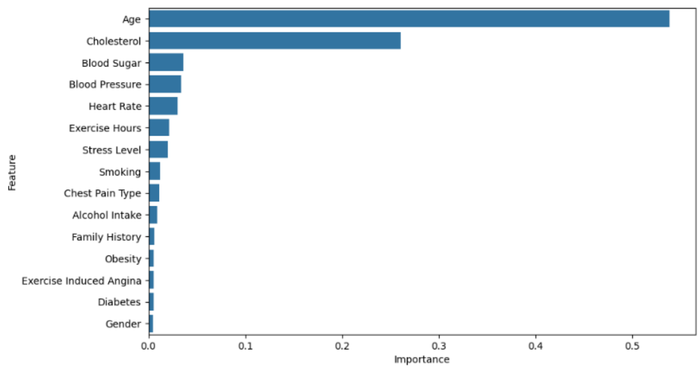
10. Feature Importance Analysis

Feature importance analysis was conducted using the Random Forest classifier to identify the most influential healthcare attributes.



	Feature	Importance
0	Age	0.532414
2	Cholesterol	0.259704
3	Blood Pressure	0.036051
12	Blood Sugar	0.034531
4	Heart Rate	0.029899
7	Exercise Hours	0.021946
11	Stress Level	0.021663
14	Chest Pain Type	0.014173
5	Smoking	0.010758
6	Alcohol Intake	0.010253
13	Exercise Induced Angina	0.006231
9	Diabetes	0.005716
10	Obesity	0.005660
8	Family History	0.005503
1	Gender	0.005499

Figure 16: Feature Importance Analysis

The analysis revealed that cholesterol level, blood pressure, blood sugar, exercise-induced angina, smoking habits, and stress levels significantly contributed to heart disease prediction.

10.1 Comparison of Clinical, Lifestyle and Demographic Factor Importance in heart disease

To determine what category of features has more predictive power on the target variable, the dataset was split into feature groups based on categories and tested individually to assess their predictive power. The dataset's features were organized into three groups: clinical/physiological (Cholesterol, Blood Pressure, Heart Rate, Blood Sugar, Chest Pain Type, Exercise Induced Angina, Diabetes, Family History, Obesity), lifestyle/behavioural (Smoking, Alcohol Intake, Exercise Hours, Stress Level), and demographic (Age, Gender). A classifier based on the Random Forest algorithm was trained on each group of features.

	Feature Set	Accuracy	Recall	ROC-AUC
0	Clinical Factors Only	0.605	0.417722	0.680877
1	Lifestyle Factors Only	0.555	0.417722	0.508892
2	Demographic Factors Only	0.845	1.000000	0.871430
3	All Factors Combined	0.995	0.987342	1.000000

It can be concluded that for the given dataset, physiological measures that are commonly taken during routine check-ups have higher predictive power than lifestyle/behavioural factors. However, in reality, behavioural patterns may contribute to a much higher degree to developing the condition over time than isolated measurements at a given point.

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11. Prediction System

A predictive healthcare system was developed to classify whether a patient is likely to suffer from heart disease based on medical and lifestyle parameters.

The trained machine learning model accepts patient details as input and generates prediction results indicating the probability of cardiovascular disease.

```
PATIENT DETAILS
   Age  Gender  Cholesterol  Blood Pressure  Heart Rate  Smoking  \
0   58       1          240             150          95        0

   Alcohol Intake  Exercise Hours  Family History  Diabetes  Obesity  \
0               1               1               1         0        1

   Stress Level  Blood Sugar  Exercise Induced Angina  Chest Pain Type
0             8          180                        1                2

Prediction Result:
Heart Disease Detected

Prediction Probability:
[[0.07 0.93]]
```

Figure 17: Sample Patient Prediction Output 1

```
PATIENT DETAILS
   Age  Gender  Cholesterol  Blood Pressure  Heart Rate  Smoking  \
0   28       1          120             120          76        0

   Alcohol Intake  Exercise Hours  Family History  Diabetes  Obesity  \
0               0               1               0         0        0

   Stress Level  Blood Sugar  Exercise Induced Angina  Chest Pain Type
0             4          110                        1                0

Prediction Result:
No Heart Disease Detected

Prediction Probability:
[[0.97 0.03]]
```

Figure 18: Sample Patient Prediction Output 2

```
PATIENT DETAILS
   Age  Gender  Cholesterol  Blood Pressure  Heart Rate  Smoking  \
0   78       0          250             180          96        0

   Alcohol Intake  Exercise Hours  Family History  Diabetes  Obesity  \
0               1               1               0         1        0

   Stress Level  Blood Sugar  Exercise Induced Angina  Chest Pain Type
0             9          160                        1                3

Prediction Result:
Heart Disease Detected

Prediction Probability:
[[0.12 0.88]]
```

Figure 19: Sample Patient Prediction Output 3

The prediction framework can assist healthcare professionals in early-stage risk assessment and preventive diagnosis.

12. Applications in Healthcare

The machine learning framework that is proposed has uses in the healthcare industry. This includes:

- finding out if someone has a disease on.
- keeping an eye on what is going on in clinics.
- checking on patients from far away.
- looking at healthcare data that is just for one person.
- trying to prevent people from getting sick.

Machine learning can be used to make models that predict things and these models can really change the healthcare industry for the better.

13. Challenges and Limitations

Even though the results are promising the proposed framework has some problems. These problems include:

- not having samples.
- making sure the results are correct in a clinical setting.
- making sure the data is real.
- dealing with data that is not balanced.
- keeping healthcare information private.

These problems make us wonder if the proposed framework really works and if it is authentic. We need to do a lot research to make the models better at predicting things in a clinical setting.

- real-world clinical validation is required.
- Data quality and imbalance may affect prediction reliability.
- Healthcare privacy and ethical concerns must be addressed.

Future research should focus on improving generalization and model explainability.

14. Future Scope

There are things that can be done in the future with this research. This includes:

- using learning to make predictive models.
- using the internet of things to monitor healthcare.
- using the cloud to look at healthcare data.
- making artificial intelligence that can be understood.
- making models that can predict many diseases at once.

The goal of this research is to make healthcare systems that are strong and reliable and that can help healthcare professionals take care of patients in the best way possible.

15. Conclusion

This study presented a machine learning framework for predicting heart disease using clinical and lifestyle data.

The framework uses data from clinics and from people’s lives. The study showed that supervised learning algorithms can be used to predict heart disease. The study used different algorithms and checked how well they worked using many different metrics.

The results show that machine learning algorithms can predict heart disease accurately. The Decision Tree and the Random Forest are the algorithms at predicting things. The results also show that using machine learning, in healthcare, have the potential to support clinical decision making and improve healthcare outcomes when used alongside medical expertise. The machine learning framework can be used to make healthcare systems that're smarter and that can help people get better care.

Code Availability

The source code, implementation details, and documentation supporting this study are available at:

https://github.com/debneil-dutt/Heart-Disease-Prediction-ML

Conflict of Interest

The author declares no conflict of interest.

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This research was independently conducted by the first author under the voluntary academic mentorship of Prof. Dr. Kunnal Sarkar. The work is not affiliated with or endorsed by Jadavpur University and does not constitute an official research project of the University.

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